A Mechanistic Greenhouse–Cucumber Digital Twin

Climate, Photosynthesis and Crop–Growth Subsystems:
Governing Equations, Variables and Control

Greenhouses Project — Model Reference

V2.2 + flags — updated 21 October 2026

Abstract

This document is a complete, self-contained specification of the mechanistic greenhouse–cucumber model that constitutes the *digital twin*. The model couples three subsystems on two time scales: a fast **climate** subsystem (a system of ordinary differential equations for the greenhouse micro-climate, integrated within each day), a **photosynthesis** subsystem (a Farquhar–von Caemmerer–Berry biochemical leaf model scaled to the canopy), and a daily **crop-growth** subsystem (a source–sink carbon-allocation map with organ pools, thermal-time development, carbohydrate-regulated flowering and fruit cohorts). Every differential equation is given with its complete right-hand side; every state variable, input and control is defined where it first appears; and the physical, biochemical and physiological principles behind each block are explained. Numerical values of all constants are collected in the appendices. Sections 2–5 specify the V2.2 base model; Section 6 documents the default-off extension flags added since (state of 21 October 2026, commit `a4cce2e`), Section 7 the best configuration found so far, Section 8 the parameter-inference scheme and Section 9 the known structural gaps. Reinforcement-learning layers are out of scope.

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1 Overview and modelling philosophy

What this section contains. A bird's-eye view of the model: the three subsystems, the variables they exchange, the two-time-scale coupling, and the notation used throughout.

1.1 The three subsystems and their coupling

The digital twin predicts how a greenhouse cucumber crop develops under a given weather series and a given control (setpoint) programme. It is organised as three coupled subsystems:

1. **Climate** (Section 2). A lumped-parameter energy- and mass-balance model of the greenhouse compartment. Its state is the vector

$$\mathbf{x}_{\text{clim}} = \big(T_1,\ T_2,\ C_1,\ V_1,\ T_5\big), \tag{1}$$

the canopy temperature T_1, the greenhouse-air temperature T_2, the air CO_2 concentration C_1, the air vapour pressure V_1, and a floor/soil thermal-mass temperature T_5. These obey the ODEs of Section 2, driven by outside weather and by the actuators.

2. **Photosynthesis** (Section 3). A biochemical model that converts the instantaneous canopy light, temperature and CO_2 into a gross assimilation rate A. It is evaluated *inside* the climate ODE (it both consumes CO_2 in the air balance and, integrated over the day, feeds the crop).

3. **Crop growth** (Section 4). A daily map that spends the day's assimilate on maintenance respiration and organ growth, advances development in thermal time, sets and grows fruit, harvests mature fruit, and updates the leaf area index (LAI).

1.2 Two time scales

The climate and photosynthesis subsystems live on a *fast* (sub-daily) scale: within one day the ODE of Section 2 is integrated over $[0, 86\,400\,\text{s}]$ at a *fixed* LAI, and the assimilation rate A is accumulated,

$$P_g \;=\; \Big(\int_0^{86\,400\,\text{s}} A(t)\,\mathrm{d}t\Big)\cdot 10^{-3}\cdot\frac{30}{44} \qquad [\text{g CH}_2\text{O m}^{-2}\,\text{d}^{-1}], \tag{2}$$

where the factor 10^{-3} converts mg $\to$ g and 30/44 converts assimilated CO_2 (molar mass 44 g/mol) to carbohydrate CH_2O (molar mass 30 g/mol). The crop-growth subsystem lives on the *slow* (daily) scale: once per day it receives P_g and the daily-mean air temperature $\overline{T_2}$, updates the crop state, and returns the new LAI, which becomes the fixed LAI for the next day's climate integration. This one-way-per-day hand-off (LAI $\to$ climate/photosynthesis $\to P_g \to$ crop $\to$ LAI) is the backbone of the twin.

1.3 Notation

Throughout, T_1, T_2, T_5 are absolute temperatures in kelvin; the crop model works with the daily-mean air temperature in °C. Symbols beginning with I (with numeric subscripts $I_1, \ldots, I_{11}$) are *inputs* to the climate right-hand side — either weather drivers or quantities supplied by the controller (Table 7); symbols $U_1, \ldots, U_{12}$ are the *control actuators* (Table 8). Greek-letter constants ($\alpha_i, \beta_i, \gamma_i, \ldots$) are fixed physical parameters listed in Appendix A. Intermediate flux quantities are named by a letter and index (h_i sensible-heat fluxes, l_i latent-heat fluxes, r_i radiative fluxes, f_i ventilation exchange rates, o_i CO_2 fluxes, p_i vapour fluxes, q_i transpiration auxiliaries); they are all defined in Section 2.

2 Climate subsystem

What this section contains. The complete micro-climate model: the five state equations (canopy temperature, air temperature, CO_2, vapour pressure and floor thermal mass), followed by the submodels that define their right-hand-side terms — shortwave (PAR/NIR) and longwave (FIR) radiation, natural ventilation, canopy transpiration and its stomatal control, condensation, and psychrometrics. The underlying principle throughout is the **lumped energy and mass balance**: each compartment (canopy, air, floor) is treated as spatially uniform, and its temperature/concentration changes at a rate equal to the net flux into it divided by its heat capacity or the compartment volume. The formulation follows the Vanthoor greenhouse model.

2.1 Governing state equations

Canopy temperature T_1. The canopy leaves store heat with capacity $\alpha_1 I_1$ per unit ground area, where $I_1 = \text{LAI}$. Their temperature responds to absorbed short-wave radiation (r_1, PAR, and r_5, NIR + lamp), long-wave exchange with the heating pipe (r_6) and with sky/floor (r_7, r_{14}), sensible-heat convection to the air (h_1) and latent-heat loss by transpiration (l_1):

$$\frac{\mathrm{d}T_1}{\mathrm{d}t} = \frac{1}{\alpha_1 I_1}\Big(r_1 + r_5 + r_6 - h_1 - l_1 - r_7 - r_{14} \Big). \tag{3}$$

Greenhouse-air temperature T_2. The air (mean height ϕ_2, density ρ_3, specific heat α_5) is heated by convection from the canopy (h_1), by the fan–pad/forced inlet (h_2), by the heating pipe (h_4) and the lamps' direct/reflector heat (h_{12}), and by absorbed radiation on building elements (r_8); it loses heat by ventilation to outside (h_7), by conduction through the soil path (h_{11}), by the net long-wave balance (r_{10}) and by fog evaporation (l_2). A separate floor-exchange term $\dot{T}_{2,\text{floor}}$ (Section 2.7) couples it to the thermal mass:

$$\frac{\mathrm{d}T_2}{\mathrm{d}t} = \frac{1}{\phi_2\,\rho_3\,\alpha_5}\Big(h_1 + h_2 + h_4 - h_7 - h_{11} + h_{12} + r_8 - r_{10} - l_2 \Big) + \dot{T}_{2,\text{floor}}. \tag{4}$$

Air CO_2 concentration C_1. Mass balance of CO_2 (in $\mathrm{mg\,m^{-3}}$) over the air column of height ϕ_2: sources are the direct air-heater combustion (o_1) and the external CO_2 dosing valve (o_2); exchange with outside occurs through the fan–pad (o_3) and through ventilation + leakage (o_5); the canopy removes CO_2 by photosynthesis at the assimilation rate $o_4 = A$:

$$\frac{\mathrm{d}C_1}{\mathrm{d}t} = \frac{1}{\phi_2}\Big(o_1 + o_2 + o_3 - o_4 - o_5 \Big). \tag{5}$$

Air vapour pressure V_1. Water-vapour balance (in Pa): the canopy adds vapour by transpiration (p_1), the fan–pad (p_2), the fog system (p_3) and the air heater (p_4); vapour is removed by ventilation (p_5), the fan–pad outflow (p_6) and condensation on the mechanical cooler (p_7), the thermal screen (p_8) and the cover (p_9). The capacity factor $\kappa_3 = \psi_1\phi_2/(\omega_2 T_2)$ converts the vapour mass balance to a pressure rate through the ideal-gas law:

$$\frac{\mathrm{d}V_1}{\mathrm{d}t} = \frac{1}{\kappa_3}\Big(p_1 + p_2 + p_3 + p_4 - p_5 - p_6 - p_7 - p_8 - p_9 \Big). \tag{6}$$

Floor / soil thermal mass T_5. See Section 2.7.

The auxiliary flux terms in Eqs. (3)–(6) are defined in Sections 2.3–2.8. We first list the transpiration auxiliaries q_i, because several balances use q_1 (the vapour-exchange coefficient) and $q_2 = \mathrm{Pws}(T_1)$ (saturated vapour pressure at leaf temperature).

2.2 Transpiration coefficient and stomatal control

Canopy transpiration is a diffusion process: water vapour flows from the saturated interior of the leaf (vapour pressure $q_2 = \mathrm{Pws}(T_1)$) to the drier greenhouse air (vapour pressure V_1), across the sum of a stomatal resistance q_3 and a fixed boundary-layer resistance. The coupling coefficient is

$$q_1 = \frac{2\,\rho_3\,\alpha_5\,I_1}{\gamma\,\gamma_2\,(\gamma_3 + q_3)}, \tag{7}$$

so that the transpiration vapour flux is $q_1(q_2 - V_1)$ and the associated latent-heat loss is $l_1 = \gamma_2 q_1 (q_2 - V_1)$, with γ_2 the latent heat of vaporisation and γ the psychrometric constant. The

stomatal resistance q_3 increases when the plant closes its stomata — in the dark, at high CO_2, and under large vapour-pressure deficit — following the Vanthoor switch functions:

$$q_7 = \big(1 + e^{(I_9-\delta_1)/\gamma_5}\big)^{-1}, \qquad q_8 = \delta_4 + (\delta_5 - \delta_4)\, q_7, \tag{8}$$

$$q_9 = \delta_6 + (\delta_7 - \delta_6)\, q_7, \qquad q_{10} = \frac{I_9 + \delta_2}{I_9 + \delta_3}, \tag{9}$$

$$q_4 = 1 + q_8\,(C_1 - 200)^2, \qquad q_5 = 1 + q_9\,(q_2 - V_1)^2, \tag{10}$$

$$q_3 = \gamma_4\, q_{10}\, q_4\, q_5. \tag{11}$$

Here I_9 is the PAR above the canopy (a radiation input, Section 2.3); q_7 is a smooth day/night switch centred at the light level δ_1; q_{10} is the radiation dependence of the minimum resistance γ_4; q_4 and q_5 inflate the resistance far from $CO_2 = 200$ ppm and at large leaf-to-air vapour deficit, respectively. The quantity $q_6 = \mathrm{Pws}(I_6)$ (saturation pressure at the mechanical-cooler temperature I_6) is used by the condensation terms.

2.3 Radiation submodel

Principle. Radiative transfer is split by wavelength band: *PAR* (photosynthetically active, 400–700 nm), *NIR* (near-infrared, solar heat), and *FIR* (far-infrared, thermal emission). Short-wave (PAR+NIR) penetration into the canopy follows the **Beer–Lambert law**: a beam is attenuated as $e^{-\beta\,\mathrm{LAI}}$ with an extinction coefficient β specific to the band. Long-wave exchange between surfaces follows the **Stefan–Boltzmann law**, net flux $\propto \sigma(T_a^4 - T_b^4)$, weighted by emissivities and view factors.

Canopy fractions. With $I_1 = \mathrm{LAI}$,

$$g_1 = 0.49\,\big(1 - e^{-\beta_3 I_1}\big), \qquad g_2 = 1 - U_1(1 - \tau_3), \qquad g_3 = \tau_1\, g_2\,\big(1 - 0.49\pi\gamma_1\phi_1\big)e^{-\beta_3 I_1}, \tag{12}$$

where g_1 is the NIR-absorbing canopy fraction, g_2 the FIR transmission through the (possibly deployed, U_1) thermal screen of transmissivity τ_3, and g_3 a combined cover–screen–canopy FIR factor.

PAR absorbed by the canopy. The transmitted outside PAR reaching the top of the canopy is $r_4 = (1-\eta_1)\tau_1\eta_2 I_2$ (I_2 = outside global radiation, η_2 its PAR fraction, τ_1 cover PAR transmissivity, η_1 the building-absorbed fraction). Direct and floor-reflected PAR absorbed by the canopy are

$$r_2 = r_4\,(1 - \rho_1)\big(1 - e^{-\beta_1 I_1}\big), \tag{13}$$

$$r_3 = r_4\, e^{-\beta_1 I_1}\,\rho_2\,(1 - \rho_1)\big(1 - e^{-\beta_2 I_1}\big), \tag{14}$$

$$r_1 = r_2 + r_3, \tag{15}$$

with ρ_1 the canopy PAR reflectance, ρ_2 the floor PAR reflectance, and β_1, β_2 the extinction coefficients for downward and floor-reflected PAR.

NIR and lamp short-wave to canopy and building.

$$r_5 = (1 - \eta_1)\,\alpha_2\,\big(\eta_3 I_2 + \eta_{14}\,\alpha_{12}\,U_{12}\big), \tag{16}$$

$$r_8 = I_2\Big(\eta_1\big(\tau_1\eta_2 + (\alpha_2 + \alpha_7)\eta_3\big) + \big(\alpha_8\eta_2 + \alpha_9\eta_3\big)\Big), \tag{17}$$

where α_2 is the canopy NIR absorptance, η_3 the NIR fraction of global radiation, $\alpha_{12}U_{12}$ the lamp radiation (U_{12} the lamp on/off actuator) and η_{14} its NIR share; r_8 collects short-wave absorbed by building/floor/cover.

Long-wave (FIR) exchange. With the sky-directed Stefan–Boltzmann kernel $B_{\text{sky}} = \epsilon_3\sigma(T_2^4 - I_4^4)$ (I_4 = sky temperature, Section 2.8),

$$r_{11} = \epsilon_4\, g_3\, \epsilon_3\, \sigma\,(I_7^4 - I_4^4), \qquad r_{12} = \epsilon_{scr}\,(U_1\tau_2)\, B_{\text{sky}}, \tag{18}$$

$$r_{13} = \epsilon_6\, B_{\text{sky}}, \qquad r_{10} = r_{11} + r_{12} + r_{13}, \tag{19}$$

$$r_{14} = \epsilon_4\,\epsilon_2\,\big(1 - 0.49\pi\gamma_1\phi_1\big)\sigma\,(T_2^4 - I_7^4), \qquad r_{15} = \epsilon_2\,\epsilon_5\,\big(1 - e^{-\beta_2 I_1}\big) g_2\,\sigma\,(T_1^4 - I_4^4). \tag{20}$$

r_{10} is the net FIR loss of the air/cover system to the sky (r_{13} through the cover of emissivity ϵ_6, r_{12} through the deployed screen of *energy-screen* emissivity ϵ_{scr}, r_{11} the floor–sky path via I_7). r_{14} is air↔floor FIR; r_{15} the canopy–sky FIR. The heating-pipe and canopy–sky canopy terms are

$$r_6 = \alpha_3\,\epsilon_1\,\epsilon_2\, g_1\,\sigma\,(I_3^4 - T_1^4), \qquad r_7 = \frac{g_1}{0.49}\,\epsilon_2\,\epsilon_3\, g_2\,\sigma\,(T_1^4 - I_4^4), \tag{21}$$

with I_3 the heating-pipe temperature and α_3, ϵ_1 the pipe area and emissivity.

2.4 Ventilation submodel

Principle. Natural ventilation exchanges air with the outside through roof and side vents. Two mechanisms drive it: *wind* (dynamic pressure $\propto I_8^2$, I_8 = wind speed) and *buoyancy* (the stack effect, $\propto g\,\Delta T/\overline{T}$). A ventilation rate is a non-negative air-exchange rate; the *direction* of any heat/CO_2/vapour transport is carried by the corresponding concentration or temperature difference in the balance, not by the rate.

Aperture-dependent rates. With U_6 the side-vent and U_8 the roof-vent apertures,

$$n_1 = \nu_1(1 - \eta_{10}U_5), \qquad n_2 = \nu_3 U_6, \qquad n_3 = \nu_2(1 - \eta_{11}U_5), \tag{22}$$

$$f_5 = \frac{n_1\, n_2\, I_8\, \sqrt{n_3}}{2\alpha_6}, \qquad f_{7t} = \max\!\Big(\omega_1\nu_6\frac{T_2 - I_5}{T_2 + I_5} + n_3 I_8^2,\ 0\Big), \qquad f_7 = \frac{U_8\,\nu_5\, n_1}{2\alpha_6}\sqrt{f_{7t}}, \tag{23}$$

where f_5 is the wind-driven side-vent exchange, f_7 the roof-vent exchange combining buoyancy (the $\omega_1\nu_6(T_2 - I_5)/(T_2 + I_5)$ stack term, $\omega_1 = g$, I_5 = outside temperature) and wind ($n_3 I_8^2$), and α_6 the floor area. The infiltration/leakage rate has a wind dependence with a floor at low wind,

$$f_6 = \begin{cases} 0.25\,\nu_4, & I_8 < 0.25 \\ \nu_4\, I_8, & I_8 \geq 0.25 \end{cases} \qquad f_{6,\text{CO2}} = \begin{cases} 0.25\,\nu_{4\text{CO2}}, & I_8 < 0.25 \\ \nu_{4\text{CO2}}\, I_8, & I_8 \geq 0.25, \end{cases} \tag{24}$$

with separate heat (ν_4) and CO_2 ($\nu_{4\text{CO2}}$) leakage coefficients. The total exchange rates entering the balances are

$$f_2 = \eta_6\, f_5 \ \ (\text{if } \eta_7 \geq \eta_8) \text{ or } \eta_6 U_1 f_5, \qquad f_3 = \frac{U_7\,\phi_8}{\alpha_6}\ (\text{forced fans}), \qquad f_4 = \eta_6\, f_7, \tag{25}$$

with η_6 the ventilation power-reduction factor and U_7 the forced-ventilation actuator.

Sensible-heat loss by ventilation.

$$h_7 \;=\; \rho_3\,\alpha_5\,\big(f_2 + f_4 + f_3 + \tfrac{1}{2}f_6\big)\,(T_2 - I_5). \tag{26}$$

The roof term f_4 is included on the same footing as the side (f_2), forced (f_3) and half-leakage ($\frac{1}{2}f_6$) terms: any air-exchange rate carries sensible heat regardless of what drives it. (Omitting f_4 here makes a calm, sunny greenhouse unable to shed its solar load.)

2.5 Sensible-heat and latent-heat fluxes of the air

$$h_1 = 2\,\alpha_4\,I_1\,(T_1 - T_2) \qquad \text{(canopy} \to \text{air convection)}, \tag{27}$$

$$h_2 = \frac{U_2\phi_7}{\alpha_6}\Big(\rho_3\alpha_5 I_5 - \gamma_2\rho_3\,\eta_5(\phi_5 - \phi_6)\Big) \qquad \text{(fan–pad inlet)}, \tag{28}$$

$$h_4 = n_{\text{pipes}}\left(1.99\,\pi\,\phi_1\,\gamma_1\,|I_3 - T_2|^{0.32}\right)(I_3 - T_2) \qquad \text{(heating pipe)}, \tag{29}$$

$$h_{11} = \frac{2\,\nu_7\,(T_2 - I_7)}{\phi_2 + \nu_8} \qquad \text{(soil-path conduction)}, \tag{30}$$

$$h_{12} = (\eta_{15} + \eta_{16})\,\alpha_{12}\,U_{12} \qquad \text{(lamp reflector + direct heat)}, \tag{31}$$

$$l_2 = \frac{\gamma_2\,U_9\,\phi_9}{\alpha_6} \qquad \text{(fog evaporation, latent)}. \tag{32}$$

h_4 is the natural-convection pipe–air exchange (a $|\Delta T|^{0.32}$ Nusselt correlation), U_2 the pad actuator, U_9 the fog actuator.

2.6 CO_2 and vapour fluxes

CO_2 fluxes ($\mathrm{mg\,m^{-2}\,s^{-1}}$):

$$o_1 = \frac{\eta_{13}\,U_4\,\lambda_4}{\alpha_6}, \qquad o_2 = \frac{U_{10}\,\psi_2}{\alpha_6}, \qquad o_3 = \frac{U_2\phi_7}{\alpha_6}(C_1 - I_{10}), \tag{33}$$

$$o_4 = A, \qquad o_5 = (C_1 - I_{10})\,\big(f_2 + f_3 + f_4 + f_{6,\mathrm{CO2}}\big), \tag{34}$$

with I_{10} the outside CO_2, U_4 the air-heater actuator, λ_4 its capacity, η_{13} the CO_2 produced per joule of combustion, U_{10} the dosing valve and ψ_2 its capacity. Vapour fluxes (Pa-rate contributions, before κ_3):

$$p_1 = q_1(q_2 - V_1), \qquad p_2 = \rho_3\frac{U_2\phi_7}{\alpha_6}\big(\eta_5(\phi_5 - \phi_6) + \phi_6\big), \tag{35}$$

$$p_3 = \frac{U_9\phi_9}{\alpha_6}, \qquad p_4 = \frac{\eta_{12}\,U_4\,\lambda_4}{\alpha_6}, \tag{36}$$

$$p_5 = \frac{\psi_1}{\omega_2}\Big(\frac{V_1}{T_2} - \frac{I_{11}}{I_5}\Big)\big(f_2 + f_3 + f_4 + f_6\big), \qquad p_6 = \frac{U_2\phi_7}{\alpha_6}\frac{\psi_1}{\omega_2}\frac{V_1}{T_2}, \tag{37}$$

with I_{11} the outside vapour pressure. Condensation occurs only when the air is supersaturated relative to the receiving surface (a one-sided switch):

$$p_7 = \begin{cases} 0, & V_1 < q_6 \\ 6.4\times10^{-9}(V_1 - q_6), & \text{else} \end{cases} \qquad \text{(cooler)}, \tag{38}$$

$$p_8 = \begin{cases} 0, & V_1 < \mathrm{Pws}(T_{\mathrm{scr}}) \\ 6.4\times10^{-9}\cdot1.7\,|T_2 - \mathrm{Pws}(T_{\mathrm{scr}})|^{0.33}(V_1 - \mathrm{Pws}(T_{\mathrm{scr}})), & \text{else} \end{cases} \qquad \text{(screen)}, \tag{39}$$

$$p_9 = \begin{cases} 0, & V_1 < \mathrm{Pws}(T_2) \\ 6.4\times10^{-9}\cdot1.2\cdot10^{3}\,|T_1 - T_2|^{0.33}(V_1 - \mathrm{Pws}(T_2)), & \text{else} \end{cases} \qquad \text{(cover)}, \tag{40}$$

where $T_{\mathrm{scr}} = \frac{1}{2}(I_5 + T_2)$ is the screen temperature.

2.7 Floor / soil thermal mass

Principle. The greenhouse floor and the top soil layer form a large *thermal buffer*: they absorb heat by day and release it at night, damping the air-temperature swing. We model this with a

single lumped floor temperature T_5 of areal heat capacity C_{soil}, exchanging with the air above (conductance k_{ground}) and with a fixed deep-soil temperature T_{deep} below (same conductance):

$$Q_{af} = k_{\text{ground}}(T_2 - T_5), \qquad Q_{fg} = k_{\text{ground}}(T_5 - T_{\text{deep}}), \tag{41}$$

$$\dot{T}_{2,\text{floor}} = -\frac{Q_{af}}{\phi_2 \rho_3 \alpha_5}, \qquad \frac{\mathrm{d}T_5}{\mathrm{d}t} = \frac{Q_{af} - Q_{fg}}{C_{\text{soil}}}. \tag{42}$$

The first equation returns the extra air-temperature rate added in Eq. (4); the second is the floor state equation. A single conductance k_{ground} (rather than a separate air–floor and floor–deep pair) is used because only the combination is identifiable from data.

2.8 Psychrometrics and sky temperature

Saturation vapour pressure (Arden Buck). For air/leaf temperature T in kelvin, with $t_c = T - 273.15$ in °C,

$$\text{Pws}(T) = \begin{cases} 611.21 \, \exp\!\left[\left(18.678 - \frac{t_c}{234.5}\right)\frac{t_c}{257.14 + t_c}\right], & t_c > 0, \\ 611.15 \, \exp\!\left[\left(23.036 - \frac{t_c}{333.7}\right)\frac{t_c}{279.82 + t_c}\right], & t_c \le 0. \end{cases} \tag{43}$$

Relative humidity and vapour-pressure deficit follow as

$$\text{RH} = 100 \, \frac{V_1}{\text{Pws}(T_2)} \ [\%], \qquad \text{VPD} = \text{Pws}(T_2) - V_1 \ [\text{Pa}]. \tag{44}$$

Effective sky temperature (Berdahl–Martin). The radiative sky temperature I_4 is colder than the air on clear nights and approaches the air temperature under overcast skies. From air temperature t_c [°C], relative humidity RH [%] and total cloud cover $c \in [0, 1]$, the dew point (Magnus) and clear-sky emissivity are

$$\gamma_m = \frac{a \, t_c}{b + t_c} + \ln \frac{\text{RH}}{100}, \quad T_{\text{dew}} = \frac{b \, \gamma_m}{a - \gamma_m}, \quad (a = 17.27, \ b = 237.7), \tag{45}$$

$$\epsilon_{\text{clear}} = \text{clip}\Big(0.711 + 0.56 \tfrac{T_{\text{dew}}}{100} + 0.73 \big(\tfrac{T_{\text{dew}}}{100}\big)^2, \ 0.6, \ 1\Big), \tag{46}$$

$$\epsilon_{\text{sky}} = \epsilon_{\text{clear}} + (1 - \epsilon_{\text{clear}}) \, c, \qquad I_4 = (t_c + 273.15) \, \epsilon_{\text{sky}}^{1/4}. \tag{47}$$

This replaces a fixed cold sky and is the dominant control on night-time radiative loss.

3 Photosynthesis subsystem

What this section contains. The biochemical leaf photosynthesis model (Farquhar–von Caemmerer–Berry, FvCB), the temperature dependence of its parameters, and the **canopy light-extinction model** that scales the leaf rate up to the whole canopy. The output is the canopy gross assimilation rate A that appears as o_4 in the CO_2 balance Eq. (5) and, integrated over the day (Eq. (2)), as the crop's carbon supply. We place the light-extinction (Beer's-law canopy) model *here*, because it is the bridge from leaf biochemistry to the canopy source term; the same extinction physics also governs the PAR/NIR absorption in the climate radiation submodel (Section 2.3).

3.1 The FvCB biochemical model

Principle. In C_3 plants (cucumber included) carbon fixation is limited by the slower of three processes: (i) the maximum rate of the Rubisco enzyme, W_c; (ii) the rate at which the light reactions regenerate RuBP, i.e. the electron-transport-limited rate W_j; and (iii) the rate at which triose phosphates are used, W_p. The net assimilation is the minimum of the three, corrected for photorespiration (through the CO_2 compensation point Γ^*) and dark respiration R_d. The inputs are leaf temperature T (here the air temperature T_2), intercellular CO_2 (taken proportional to the air CO_2 C_1) and absorbed irradiance (from the canopy PAR I_9).

Inputs and conversions. With incident radiation I_w [W m^{-2}] and air CO$_2$ C [mg m^{-3}],

$$I = 4.6 \times 10^{-6}\, I_w \quad [\text{mol photons m}^{-2}\,\text{s}^{-1}], \qquad C_i = 0.509 \times 10^{-6}\, C \quad [\text{mol m}^{-3}]. \tag{48}$$

CO_2 compensation point and electron transport.

$$\Gamma^* = \frac{0.5\, O_a}{S_{c/o,25}\, \exp\!\left[\frac{T-298.15}{T} \cdot \frac{E_{S_{c/o}}}{298.15\, R}\right]}, \tag{49}$$

$$J = \frac{(\alpha I + J_{\max}) - \sqrt{(\alpha I + J_{\max})^2 - 4\,\theta\,\alpha I\, J_{\max}}}{2\,\theta}, \tag{50}$$

where O_a is the O$_2$ mole fraction, $S_{c/o,25}$ the Rubisco specificity and $E_{S_{c/o}}$ its activation energy; α is the initial quantum-yield slope, $J_{\max}$ the maximum electron transport rate and θ the curvature of the non-rectangular hyperbola relating J to light.

Temperature-adjusted Rubisco kinetics.

$$V_{c\max} = V_{c\max,25}\, \frac{Q_{10,Vc}^{0.1(T-298.15)}}{1 + \exp[0.128\,(T - 315.15)]}, \tag{51}$$

$$K_C = K_{C,25}\, Q_{10,KC}^{0.1(T-298.15)}, \qquad K_O = K_{O,25}\, Q_{10,KO}^{0.1(T-298.15)}, \tag{52}$$

with a peaked (deactivating) temperature response for $V_{c\max}$ and Michaelis constants K_C, K_O for CO$_2$ and O$_2$. The effective (stomatal+mesophyll) conductance is held at the constant Bush (2023) value

$$g_{\text{eff}} = \frac{0.3 \cdot 0.25}{0.3 + 0.25} \approx 0.1364 \quad [\text{mol m}^{-2}\,\text{s}^{-1}]. \tag{53}$$

The three limiting rates. Each limitation gives a quadratic in the assimilation rate A_r; the physically admissible (lower) root is taken:

$$\begin{aligned}
\text{Rubisco:}\quad & p_c = -\big(V_{c\max} + g_{\text{eff}}(C_i + K_C(1 + O_a/K_O)) - R_d\big),\\
& q_c = g_{\text{eff}}\big(V_{c\max}\max(C_i - \Gamma^*, 0) - (C_i + K_C(1 + O_a/K_O))R_d\big),\\
& A_{r,c} = \tfrac{1}{2}\big(-p_c - \sqrt{p_c^2 - 4q_c}\big),
\end{aligned} \tag{54}$$

$$\begin{aligned}
\text{Light:}\quad & p_j = -0.25\, J + R_d - g_{\text{eff}}(C_i + 2\Gamma^*),\\
& q_j = 0.25\, g_{\text{eff}}\max(C_i - \Gamma^*, 0)\, J - g_{\text{eff}}(C_i + 2\Gamma^*)R_d,\\
& A_{r,j} = \tfrac{1}{2}\big(-p_j - \sqrt{p_j^2 - 4q_j}\big),
\end{aligned} \tag{55}$$

$$\text{Product:}\quad A_{r,p} = 1.5\, V_{c\max} - R_d. \tag{56}$$

The **leaf-level** net assimilation (per unit *leaf* area) is $A_{\text{leaf}} = \min(A_{r,j}, A_{r,c}, A_{r,p})$, and the **canopy** assimilation returned to the balances is

$$A \;=\; 44010 \;\cdot\; L_{\text{eff}} \;\cdot\; \min(A_{r,j}, A_{r,c}, A_{r,p}) \quad [\text{mg CO}_2\,\text{m}^{-2}\,\text{s}^{-1}], \tag{57}$$

where the factor 44010 converts mol→mg of CO$_2$, and L_{eff} is the light-intercepting effective LAI defined next. For the cucumber crop under CO$_2$ enrichment the canopy operates predominantly in the *light-limited* regime ($A_{r,j}$ smallest), so radiation — solar plus supplemental lamp PAR — is the limiting resource.

3.2 Canopy light-extinction model

Principle (Beer–Lambert). Leaves higher in the canopy shade those below, so irradiance decays with cumulative leaf area ℓ (LAI measured from the top) as $I(\ell) = I_0\, e^{-k\ell}$, with k the canopy extinction coefficient ($k \approx 0.7$ for cucumber). Two canopy-scaling schemes are used.

(a) Big-leaf (effective-LAI) scaling. The leaf rate is evaluated once at the top-of-canopy light I_0 and multiplied by the *light-intercepting effective LAI*

$$L_{\text{eff}}(\text{LAI}) \;=\; \frac{1 - e^{-k\,\text{LAI}}}{k}, \tag{58}$$

which is the vertical integral of the interception profile and *saturates* at $1/k$ as LAI$\to\infty$ (extra leaf intercepts almost no additional light). This L_{eff} is the factor used in Eq. (57) and, with the corresponding $\beta_1, \beta_2, \beta_3$ coefficients, underlies the PAR/NIR absorption of Section 2.3.

(b) Layered (vertical-profile) scaling. A more faithful scheme integrates the *leaf* rate down the light gradient. Discretising the canopy into leaf layers of area a_i at cumulative depth ℓ_i (mid-layer),

$$A \;=\; 44010 \sum_i a_i \, A_{\text{leaf}}\big(I_0\, e^{-k\,\ell_i}\big). \tag{59}$$

When the leaf response is linear in light (the light-limited regime) Eq. (59) reduces exactly to the big-leaf form (57)–(58); when upper leaves are light-saturated it correctly predicts a *higher* canopy assimilation, because shaded lower leaves operate at their more efficient (steeper) part of the light-response curve. Scheme (b) is the one used by the leaf-cohort crop model (Section 4.7), where the layers are the leaf-age cohorts; scheme (a) is used by the single-pool crop model (Section 4.6).

4 Crop-growth subsystem

What this section contains. The daily crop model: how the day's assimilate P_g is partitioned into maintenance, organ growth and fruit; how development proceeds in thermal time; the **flowering / fruit-set equations**; fruit growth and harvest; and the leaf-area update. Two variants share the same physiology and differ only in how the canopy is represented: a *single-pool* model (Section 4.6) and a *leaf-age-cohort / vertical-canopy* model (Section 4.7). The **planting density** and its role are defined in Section 4.8. The governing biological principle is **source–sink carbon partitioning**: a common assimilate pool (the source) is distributed among competing organs (the sinks) in proportion to their potential growth (sink strength), limited by supply when the source is scarce.

4.1 State variables and thermal time

The single-pool crop state is

$$\big(W_{\text{leaf}},\; W_{\text{stem}},\; W_{\text{root}},\; \text{LAI},\; \{\text{fruit cohorts}\},\; Y_{DW},\; Y_{FW},\; n_{\text{node}}\big), \tag{60}$$

with organ dry weights $W_\bullet$ [g m^{-2}], leaf area index LAI, a list of fruit cohorts, the cumulative harvested fruit dry (Y_{DW}, g m^{-2}) and fresh (Y_{FW}, kg m^{-2}) weight, and the main-stem node number n_{node}. Each **fruit cohort** carries a development stage dev $\in [0, 1]$ (its thermal age), a dry weight W [g m^{-2}] and a number of fruits n [fruits m^{-2}].

Thermal time. Development is driven by temperature above a base T_{base} (**degree-days**): with daily-mean air temperature $\overline{T_2}$ [°C],

$$\Delta T \;=\; \max\big(\overline{T_2} - T_{\text{base}},\, 0\big). \tag{61}$$

Nodes appear on the growing head at rate r_{node} until *topping* on day d_{top}, and each fruit cohort ages by $\Delta\text{dev} = \Delta T/\text{DD}_{\text{fruit}}$ per day, where DD_{fruit} is the degree-days from set to ripe:

$$n_{\text{node}} \mathrel{+}= r_{\text{node}}\,\Delta T \;\; (\text{if } d < d_{\text{top}}), \qquad \text{dev} \mathrel{+}= \frac{\Delta T}{\text{DD}_{\text{fruit}}} \;\; (\text{all cohorts}). \tag{62}$$

4.2 Respiration, supply and growth conversion

Maintenance respiration. Living tissue consumes carbohydrate to maintain itself at a rate that doubles roughly every 10 °C (a Q_{10} law):

$$q_{10} = Q_{10,\mathrm{resp}}^{(\overline{T_2}-25)/10}, \tag{63}$$

$$R_m = q_{10}\big(k_{m,\mathrm{leaf}}W_{\mathrm{leaf}} + k_{m,\mathrm{stem}}W_{\mathrm{stem}} + k_{m,\mathrm{root}}W_{\mathrm{root}} + k_{m,\mathrm{fruit}}W_{\mathrm{fruit}}\big), \tag{64}$$

with organ-specific maintenance coefficients $k_{m,\bullet}$ [g CH_2O g^{-1} d^{-1}] and $W_{\mathrm{fruit}} = \sum_{\mathrm{cohorts}} W$.

Assimilate available for growth. What remains of the day's gross assimilate after maintenance is converted to structural dry matter with a growth (conversion) efficiency 1/asrq:

$$C_{\mathrm{av}} = \max(P_g - R_m,\, 0), \qquad \Delta\mathrm{DM}_{\mathrm{pot}} = \frac{C_{\mathrm{av}}}{\mathrm{asrq}} \quad [\mathrm{g\,DM\,m^{-2}\,d^{-1}}], \tag{65}$$

where asrq (assimilate required per unit dry matter, > 1) accounts for growth respiration. $\Delta\mathrm{DM}_{\mathrm{pot}}$ is the potential (source-side) dry-matter increment.

4.3 Sinks, source/sink limitation and allocation

Sink strengths. The vegetative sink fills the canopy toward a density-dependent cap L_{cap} and switches off as the cap is approached; each fruit cohort's sink follows a bell-shaped growth curve. With the density factor $\mathrm{d} = \mathrm{stem_density}/\mathrm{d}_{\mathrm{ref}}$ (Section 4.8),

$$L_{\mathrm{cap}} = \mathrm{LAI}_{\max}\,\mathrm{d}, \qquad S_{\mathrm{veg}} = \max\Big(\mathrm{veg_sink}_{\max}\,\mathrm{d}\,\big(1 - \tfrac{\mathrm{LAI}}{L_{\mathrm{cap}}}\big),\, 0\Big), \tag{66}$$

$$S_{\mathrm{fruit},i} = n_i\, W_{f\,\max}\, \varsigma(\mathrm{dev}_i)\,\Delta\mathrm{dev}, \qquad \varsigma(\mathrm{dev}) = 6\,\mathrm{dev}\,(1-\mathrm{dev}), \tag{67}$$

where $W_{f\,\max}$ is the potential mature fruit dry weight and the *sink-shape* $\varsigma(\mathrm{dev}) = 6\,\mathrm{dev}(1-\mathrm{dev})$ (the derivative of the smoothstep $\mathrm{dev}^2(3-2\mathrm{dev})$) integrates to unity over $\mathrm{dev} \in [0,1]$, so a cohort's lifetime potential growth totals $n\,W_{f\,\max}$.

Allocation. The total demand is $S_{\mathrm{tot}} = S_{\mathrm{veg}} + \sum_i S_{\mathrm{fruit},i}$. The realised increment is *source-limited* when supply is scarce and *sink-limited* when supply is ample:

$$\Delta\mathrm{DM} = \min\big(\Delta\mathrm{DM}_{\mathrm{pot}},\, S_{\mathrm{tot}}\big), \tag{68}$$

and is split among organs in proportion to their sink share:

$$\Delta\mathrm{DM}_{\mathrm{veg}} = \Delta\mathrm{DM}\,\frac{S_{\mathrm{veg}}}{S_{\mathrm{tot}}}, \qquad W_{\mathrm{leaf}} \mathrel{+}= f_{\mathrm{leaf}}\,\Delta\mathrm{DM}_{\mathrm{veg}}, \tag{69}$$

$$W_{\mathrm{stem}} \mathrel{+}= f_{\mathrm{stem}}\,\Delta\mathrm{DM}_{\mathrm{veg}}, \qquad W_{\mathrm{root}} \mathrel{+}= f_{\mathrm{root}}\,\Delta\mathrm{DM}_{\mathrm{veg}}, \tag{70}$$

$$W_i \mathrel{+}= \Delta\mathrm{DM}\,\frac{S_{\mathrm{fruit},i}}{S_{\mathrm{tot}}} \quad \text{(each fruit cohort)}, \tag{71}$$

with vegetative partitioning fractions $f_{\mathrm{leaf}} + f_{\mathrm{stem}} + f_{\mathrm{root}} = 1$.

4.4 Flowering: carbohydrate-regulated fruit set

Principle. New fruits are set continuously once the plant is old enough ($d \geq d_{\mathrm{set,start}}$), but the *rate* of setting is regulated by the plant's carbohydrate status: when the standing fruit load already demands more carbon than the source can supply, further set is suppressed (young fruit abort / fail to set); once those fruits are harvested and the load drops, set resumes. Combined

with the ∼2-week fruit-growth delay ($\mathrm{DD}_{\mathrm{fruit}}$), this supply/demand feedback produces the characteristic weekly harvest *flushes*. Define the instantaneous supply/demand ratio

$$s \;=\; \frac{\Delta \mathrm{DM}_{\mathrm{pot}}}{S_{\mathrm{tot,pre}} + 10^{-9}}, \qquad S_{\mathrm{tot,pre}} = S_{\mathrm{veg}} + \textstyle\sum_i S_{\mathrm{fruit},i}, \tag{72}$$

evaluated *before* the new cohort is added. The set fraction is a clipped linear ramp between a lower and upper carbohydrate threshold,

$$f_{\mathrm{set}} \;=\; \mathrm{clip}\Big(\frac{s - s_{\mathrm{low}}}{s_{\mathrm{high}} - s_{\mathrm{low}}},\, 0,\, 1\Big), \tag{73}$$

so that below s_{low} no fruit sets and above s_{high} set is maximal. On each day with $d_{\mathrm{set,start}} \le d < d_{\mathrm{top}}$ a new fruit cohort is created with number

$$n_{\mathrm{set}} \;=\; \mathrm{set_rate} \cdot \mathrm{d} \cdot f_{\mathrm{set}} \qquad \text{(single-pool model)}, \tag{74}$$

where set_rate is the potential fruits per unit area per day and d the density factor. (In the cohort model, Section 4.7, set is driven by node appearance rather than calendar day; see Eq. (78).)

4.5 Fruit growth, harvest and leaf-area update

A cohort is harvested when it reaches dev ≥ 1; its dry weight is banked and converted to fresh weight through the fruit dry-matter content $\mathrm{DMC}_{\mathrm{fruit}}$:

$$Y_{DW} \mathrel{+}= W, \qquad Y_{FW} \mathrel{+}= \frac{W/\mathrm{DMC}_{\mathrm{fruit}}}{1000} \;\; [\mathrm{kg\,FW\,m^{-2}}]. \tag{75}$$

Finally the leaf area is refreshed from the leaf dry weight through the specific leaf area (SLA, leaf area per unit leaf dry weight) and capped (see pruning, Section 5.9):

$$\mathrm{LAI} = \mathrm{SLA} \cdot W_{\mathrm{leaf}}. \tag{76}$$

4.6 Single-pool canopy model

The model of Sections 4.1–4.5 *is* the single-pool model: leaves form one lumped pool W_{leaf}, the canopy source uses the big-leaf light scaling (Eq. (58)), and pruning is a cap on total leaf area (Section 5.9). It is compact and fast, but cannot represent the vertical age/light structure of a real high-wire cucumber canopy — which motivates the cohort model.

4.7 Leaf-age-cohort (vertical-canopy) model

Principle. A high-wire cucumber produces new leaves at the head while old leaves at the base become deeply shaded, senesce, and are periodically removed by the grower. The cohort model represents the canopy as a stack of **leaf-age cohorts**, each with an area and a thermal age, so that (i) photosynthesis is integrated layer-by-layer down the light gradient (Eq. (59)), (ii) each cohort's maintenance respiration is charged on its own mass, and (iii) de-leafing removes the oldest, most-shaded cohorts. This yields a physically meaningful optimal LAI and a faithful account of de-leafing that the single pool cannot provide.

Leaf production (node-driven). Rather than filling a cap, leaf area is produced in step with node appearance. With leaf area demanded per node ℓ_{node} and $\Delta n_{\mathrm{node}} = r_{\mathrm{node}} \Delta T$ the day's new nodes,

$$\text{leaf-area demand} = \ell_{\mathrm{node}}\, \mathrm{d}\, \Delta n_{\mathrm{node}}, \qquad S_{\mathrm{veg}} = \frac{\ell_{\mathrm{node}}\, \mathrm{d}\, \Delta n_{\mathrm{node}}/\mathrm{SLA}}{f_{\mathrm{leaf}}}, \tag{77}$$

the new top cohort receiving $f_{\mathrm{leaf}}\, \Delta \mathrm{DM}_{\mathrm{veg}}\, \mathrm{SLA}$ of area.

Node-driven flowering. Fruit set is triggered per node flush rather than per day, with the same carbohydrate regulation f_{set} of Eq. (73):

$$n_{\text{set}} = \text{set_rate} \cdot \text{d} \cdot \Delta n_{\text{node}} \cdot f_{\text{set}} \qquad \text{(cohort model)}. \tag{78}$$

Senescence and de-leafing. A cohort is dropped by natural senescence once its thermal age exceeds the leaf lifespan $\text{life}_{\text{leaf}}$ [degree-days]. Management de-leafing removes the *oldest* cohorts down to a working target LAI L_{deleaf}: with $\text{LAI} = \sum_i a_i$, cohorts are trimmed oldest-first until $\text{LAI} \le L_{\text{deleaf}}$. Because the removed leaves are the deepest and most shaded, they contribute almost no assimilation (Eq. (59)) yet still respired, so their removal is close to cost-free in carbon and sheds their maintenance load.

4.8 Planting density

Where it lives and what it does. Planting density (stems per m^2) is a *crop-structural / management* parameter, so it is defined here, in the growth subsystem; it is also a control decision, discussed in Section 5.9. All the per-area capacities of the growth model (the vegetative sink $\text{veg_sink}_{\max}$, the fruit-set rate set_rate, the leaf-area cap $\text{LAI}_{\max}$ and the per-node leaf demand ℓ_{node}) are defined at a *reference* density d_{ref} and scale linearly with the actual density through the factor

$$\text{d} \;=\; \frac{\text{stem_density}}{\text{d}_{\text{ref}}}, \tag{79}$$

which appears in the sink and set equations above. At $\text{stem_density} = \text{d}_{\text{ref}}$ the factor is unity, so all reference calibrations are recovered exactly. A denser stand closes the canopy (raises LAI) faster and intercepts light earlier; because interception saturates (Eq. (58)) while maintenance respiration keeps growing with leaf mass, there is a finite *optimal* density, and its value is governed primarily by the light-extinction coefficient k (Section 3.2). The initial canopy is also scaled by density, $\text{LAI}_0 \leftarrow \text{LAI}_0\,\text{d}$.

5 Control variables and management strategies

What this section contains. The actuators that drive the model, the setpoint programme that a grower (or an automatic controller) specifies, the low-level feedback controllers that translate setpoints into actuator positions, and the discrete crop-management strategies (pruning/de-leafing, topping, density scheduling). The controlled variable is the greenhouse *air* temperature T_2 (the physically measured quantity), not the canopy temperature.

5.1 Actuators

Twelve actuators $U_1, \ldots, U_{12}$ enter the climate right-hand side (Table 8). In the closed-loop twin only a subset is active: the thermal/energy screen U_1, the side vent U_6, the roof vent U_8, the fog U_9, the CO_2 valve U_{10} and the lamps U_{12}; the heating pipe enters as its temperature I_3. The pad/heater/forced-ventilation actuators (U_2, U_3, U_4, U_5, U_7) are available but set to zero in the reference configuration.

5.2 Setpoint programme (the action surface)

A control programme is a set of within-day setpoint schedules:

- day and night **air-temperature** setpoints T_{set} (a day/night step schedule);
- **CO_2** setpoint C_{set};
- **lamp** on/off schedule (start hour, duration ⇒ end hour);

- ventilation **proportional band** p_{band} and dead-band *offset* above the temperature setpoint;
- the outside-temperature threshold $T_{\text{out,max}}$ (screen-closing) and radiation threshold Rad_{scr};
- the **VPD floor** VPD_{min} for humidity control.

5.3 Heating controller (PI → pipe temperature)

A leaky-integral proportional–integral controller on the temperature error $e_T = T_{\text{set}} - T_2$ sets a heating fraction $U_{\text{heat}} \in [0, 1]$, which blends the maximum pipe temperature $T_{\text{pipe}}^{\max}$ with the air temperature to give the pipe temperature I_3:

$$U_{\text{heat}} = \text{clip}\big(K_p^T e_T + K_i^T e_T^{\text{int}},\, 0,\, 1\big), \qquad I_3 = U_{\text{heat}}\, T_{\text{pipe}}^{\max} + (1 - U_{\text{heat}})\, T_2, \tag{80}$$

where the integral state obeys $\dot{e}_T^{\text{int}} = e_T - \lambda_T e_T^{\text{int}}$ (a *leaky* integrator with decay λ_T), *frozen* under actuator saturation (anti-windup, Section 5.10). I_3 then drives the pipe heat flux h_4 (Section 2.5) and the pipe FIR r_6.

5.4 CO_2 dosing controller (PI → valve)

Analogously, on the error $e_C = C_{\text{set}} - C_1$,

$$U_{10} = \text{clip}\big(K_p^C e_C + K_i^C e_C^{\text{int}},\, 0,\, 1\big), \tag{81}$$

opening the external CO_2 valve (source $o_2 = U_{10}\psi_2/\alpha_6$), with the same leaky-integral and anti-windup treatment.

5.5 Ventilation controller (proportional band → side/roof)

Ventilation opens on a proportional band above the temperature setpoint plus a dead-band offset:

$$U_{\text{vent}} = \begin{cases} 0, & T_2 < T_{\text{set}} + \text{ofs}, \\ \dfrac{T_2 - T_{\text{set}} - \text{ofs}}{p_{\text{band}}}, & T_{\text{set}} + \text{ofs} \le T_2 < T_{\text{set}} + \text{ofs} + p_{\text{band}}, \\ 1, & \text{otherwise}, \end{cases} \tag{82}$$

split so the side vent opens first and the roof vent follows:

$$U_6 = 2\min(U_{\text{vent}}, 0.5), \qquad U_8 = 2\max(0,\, U_{\text{vent}} - 0.5). \tag{83}$$

5.6 Thermal / energy-screen controller

The screen is deployed to retain heat when it is dark (radiation $I_2 < \text{Rad}_{\text{scr}}$) or cold outside ($I_5 < T_{\text{out,max}}$), and is cracked open to shed moisture when the air is too humid (VPD below the floor):

$$U_1 = \begin{cases} 0.85, & \text{closed and VPD} < \text{VPD}_{\text{min}} \text{ (crack to dehumidify)}, \\ 1, & \text{closed (dark or cold) and not too humid}, \\ 0, & \text{open (day)}. \end{cases} \tag{84}$$

The screen's FIR emissivity is the low-emissivity *energy-screen* value ϵ_{scr} (Section 2.3), distinct from a full-emissivity shade screen, so that deploying it genuinely retains long-wave heat.

5.7 Humidity control by vapour-pressure deficit

Humidity is regulated on the **VPD** (Section 2.8), not relative humidity, because VPD is the physiologically and physically meaningful driver of transpiration and condensation and is temperature-independent. When $\mathrm{VPD} = (\mathrm{Pws}(T_2) - V_1)/1000$ [kPa] falls below the floor $\mathrm{VPD}_{\min}$ (air too close to saturation, condensation/disease risk), a proportional dehumidification demand is added to the vents (and the screen is cracked as above):

$$U_{\mathrm{hv}} = \mathrm{clip}\Big(\frac{\mathrm{VPD}_{\min} - \mathrm{VPD}}{\mathrm{band}},\, 0,\, 1\Big)\, v_{\max}, \qquad U_6 \leftarrow \max(U_6, U_{\mathrm{hv}}),\quad U_8 \leftarrow \max(U_8, U_{\mathrm{hv}}). \tag{85}$$

5.8 Supplemental lighting

The lamps contribute PAR to photosynthesis (added to the canopy light I_w in the layered source, Section 3.2), NIR heat to the canopy (r_5, via η_{14}), and reflector/direct heat to the air (h_{12}, via η_{15}, η_{16}), at an electrical intensity α_{12} per unit area when on ($U_{12} = 1$). Because lamp electricity is the dominant operating cost and its value depends on the free solar light available and on time-of-use tariffs, the lamp *schedule* (start hour and duration) is a primary management lever.

5.9 Crop-management strategies

Pruning / de-leafing. Leaf area is limited to the smaller of the density cap L_{cap} and a management de-leaf target L_{deleaf}. In the single-pool model this caps W_{leaf} (Eq. (76)); in the cohort model it removes the oldest, most-shaded cohorts down to L_{deleaf} (Section 4.7). De-leafing sheds the maintenance-respiration burden of leaves that intercept little light, without touching stems or fruit.

Topping. Node appearance (vegetative growth of the head) stops at the topping day d_{top}, after which the plant only ripens its existing fruit load.

Density scheduling (interplanting / plant separation). The planting density (Section 4.8) may be changed during the season: planted densely for fast early canopy closure, then thinned (stems pulled) later. A density reduction on a given day scales the standing leaf area and the standing (unharvested) fruit load by the ratio of new-to-old density (the pulled stems leave with their leaves and green fruit), and all subsequent per-area production runs at the reduced density.

5.10 Anti-windup

Because the PI integrators are ODE states, they can *wind up* while an actuator is saturated. The integral is therefore frozen (its accumulation term set to the leak only) whenever the actuator is at a bound *and* the error would push it further into saturation:

$$\dot{e}^{\mathrm{int}} = \begin{cases} -\lambda\, e^{\mathrm{int}}, & \text{saturated in the driving direction,} \\ e - \lambda\, e^{\mathrm{int}}, & \text{otherwise,} \end{cases} \tag{86}$$

for both the temperature (e_T, λ_T) and CO_2 (e_C, λ_C) loops. This prevents the runaway that windup causes once the leakage $\nu_4, \nu_{4\mathrm{CO2}}$ is set to its physical value.

6 The model as coded: V2.2 with extension flags (state of 21 October 2026)

What this section contains. Sections 2–5 describe the *V2.2 base* (all extension flags off). Since September the engine has been extended under a strict rule: **the V2.2 engine is frozen**

and every new process enters as a default-off flag, so that with all flags at their JSON defaults the code reproduces V2.2 bit for bit. This section lists every flag, the equation it changes and the setting used by the current recommended configuration ("n79"; Section 7). It describes git commit `a4cce2e` of the `invernaderos` repository. Notes "Note N" refer to the numbered project notes.

6.1 Code map and state vector

file (`invernaderosV2.2/GreenhouseSim/`)	content
`src/climate.jl`	psychrometrics, algebraic cover node (`cover_T`, `cover_latent`), `floor_full_fluxes`, `climate_rhs` ($\dot{T}_1, \dot{T}_2, \dot{C}_1, \dot{V}_1$)
`src/forward_map.jl`	`rhs!`: controls, controllers, optional states, lamps, internal mass, floor, crop respiration; `phys_par`
`src/screen_optics.jl`	roof–screens–canopy–floor solar column (`solar_column`, `SolarOut`)
`src/photosynthesis.jl`	FvCB canopy assimilation
`src/growth.jl`, `generic_growth.jl`	V2.2 crop and the generic M94 path; Aaron's canopy step
`src/measured.jl`	measured-climate drivers for the crop calibration
parameters	`invernaderos2.0/code/configfiles/constants_climate.json`, `constants_cropphoto.json`; `GreenhouseSim/configfiles/cucumber_growth.json`

The ODE state (per m^2 floor) is

$$u = \big[\, T_1,\ T_2,\ C_1,\ V_1,\ e^{\mathrm{int}},\ T_5,\ U_8^s,\ T_p^s,\ T_d,\ T_m \,\big], \tag{87}$$

with canopy, air, CO_2, vapour pressure, heating-error integral and floor layer always present; U_8^s (vent state, `vent_ctrl`) and T_p^s (lagged pipe, `pipe_lag`) exist when length$(u) \geq 8$, the deep soil T_d (`soil_dyn`) when ≥ 9 and the internal mass T_m (`mass_node`) when ≥ 10. The cover has no state (algebraic node) and the screens have no temperature ($T_{\mathrm{scr}} = \frac{1}{2}(T_{\mathrm{out}} + T_2)$). Each day is integrated separately (Tsit5) and the end state carried over.

6.2 Known quirks of the V2.2 base

V2.2 is V2.0-faithful apart from three non-flag fixes (non-negative ventilation rate, roof vent f_4 in the sensible-heat loss h_7, and the identifiable floor coupling k_{ground}). It keeps these V2.0 features, each of which a flag corrects:

- photosynthesis is evaluated with "LAI" $= I_9$ (global radiation) $\rightarrow$ `photo_lai`;
- the stomatal resistance swaps the day/night coefficients and reads $\mathrm{mg\,m^{-3}}$ as ppm $\rightarrow$ `stomata_fix`;
- the air receives $(\alpha_8\eta_2 + \alpha_9\eta_3)I_2 = 1.0\,I_2$ as "cover absorption" on top of the canopy share $\rightarrow$ `solar_split`, `solar_cons`;
- the canopy absorbs a fixed $\alpha_2 = 0.35$ of the outside NIR, independent of LAI (0.35 is Vanthoor's *reflection*) $\rightarrow$ `solar_cons`;
- the lamps release only $\approx 29\,\%$ of their electrical power as heat $\rightarrow$ `lamp_full`;
- screen condensation has no U_1 factor and mixes K with Pa $\rightarrow$ `vapour_fix`.

6.3 Flag catalogue (climate engine)

JSON = default in `constants_climate.json` (V2.2 behaviour except where marked *); n79 = value in the recommended configuration.

flag (parameters)	JSON	Note	effect	n79
`use_cover_node` (h_{out})	1*	29	algebraic cover temperature, eq. (88), replaces T_{sky} in $r_7, r_{11}, r_{12}, r_{13}, r_{15}$	on; h_{out} inferred
`nu4_min`, `U8_min_leak`	0	–	floors on leakage and vent opening	0
`heat_ctrl` ($K_p, K_i, T_{\max}, \Delta T_h$)	0	33–34	PI heating, eq. (90)	on (10, 10/1800, 65 °C, 0)
`hc_log_pipe`	0	42	logged pipe $> T_2 + 1$ K heats alone; PI elsewhere	on
`pipe_lag` (τ_p)	0	34	PI pipe as lagged state T_p^s	off
`vent_ctrl` (g, e_0, τ_v, h, b)	0	34–37	closed-loop roof vents, eq. (91)	off (open loop)
`vc_night_log`, `vc_min_night`, `vc_min_log`	0	38–42	night replay / minimum openings (only with `vent_ctrl`)	min_log on (no-op)
`vent_exp`	1	33	effective opening $U_8^{\text{vent_exp}}$	1
`vent_screen` (p_{scr})	0	33	$U_8 \to U_8\,[1 - U_1(1 - p_{\text{scr}})]$: a closed screen throttles roof ventilation of the crop air	on; p_{scr} inferred
`vent_split` (e, g_w, c_l, c_w)	0	58	lee and wind vents as two flows, eq. (92)	on
`black_scr` (k_b)	0	34	$U_1^{\text{eff}} = 1 - (1 - U_1)(1 - k_b\,U_{\text{black}})$	off
`stomata_fix`	0	33	day coefficients weighted by daylight; C_1 in ppm ($C_1 T_2/536.4$)	on
`vapour_fix`	0	33	$p_7 \times U_3$; $p_8 = 6.4\times10^{-9}\,1.7\,U_1\lvert T_2 - T_{\text{scr}}\rvert^{0.33}(V_1 - P_s(T_{\text{scr}}))$	on
`cond_heat` (f_c)	0	33	$+f_c\gamma_2 p_8$ to the air	on, $f_c = 0$
`dew_bl`	0	85	if $P_s(T_1) < V_1$: q_1 with boundary layer γ_3 only	on
`soil_dyn` (τ_d)	0	43	$\dot{T}_d = (T_5 - T_d)/\tau_d$ replaces fixed T_{soil}	on, 14 d
`floor_full`	0	46	floor balance closed: r_{11} from floor, r_{14} with T_1, h_{11} to floor	on
`solar_split` (a_c)	0	46	drop the $1.0\,I_2$ to the air; $a_c I_2$ to cover; transmitted sun to the floor	on
`deep_k` (k_{deep})	0	46	own floor→deep conductance	on
`mass_node` (C_m, h_m, f_m)	0	48	internal mass, eq. (93)	on
`lamp_full` ($P_{\text{el}}, f_{\text{can}}$)	0	54	full lamp power, eq. (94)	on, $P_{\text{el}} = 117$
`lamp_scr_loss`	0	58d	lamp share lost through roof with open screens	0
`air_cap_mult`	1	61	$\dot{T}_2 \to \dot{T}_2/m_{\text{air}}$ (fast mass lumped with air)	inferred
`s_rad`, `dday`	0	50, 61	measurement terms in the calibration script only	removed
`phys_light` (P_{lamp})	1*	71	physiology PAR = sun through screens + lamp PAR ($40.65\,\text{W}\,\text{m}^{-2}$)	on
`photo_lai`	1*	71	assimilation $\times(1 - e^{-0.7L})/0.7$ instead of $L = I_9$	on
`cover_area`	1*	79	$r_{13} \times A_c$ (cover m^2 per floor m^2)	1.6
`cover_cond`	0	79, 81	condensation on the cover, eq. (89); latent heat warms the cover	on
`co2_resp`	0	79	$\dot{C}_1$ += $R_{\text{crop}}(t)\,2^{(T_1 - 293.15)/10}/\phi_2$	on
`leak_shared`	0	86	CO_2 leakage uses ν_4	on
`rd_q10`	0	86	$R_d \times 2^{(T_1 - 298.15)/10}$	on
`leak_stack` (c_s)	0	88	$f_6 = \nu_4\sqrt{\max(v, 0.25)^2 + c_s\lvert T_2 - T_{\text{out}}\rvert}$; full f_6 in h_7 (V2.2: $\frac{1}{2}$)	on
`crop_cap` ($c_{p,\text{crop}}$)	0	91	$C_m \to C_m + c_{p,\text{crop}}\,\text{FW}(t)$	on, 3800
`scr_solar`	0	101	screen optics factors on the V2.2 solar terms (superseded)	off

flag (parameters)	JSON	Note	effect	n79
`solar_cons` (optics)	0	103	energy-conserving solar column, Section 6.5	on
`solar_mass_share`	0	103	mass takes f_m of the floor-level solar	on
`par_below`	0	105	physiology sees PAR on both canopy faces	on

Script-level switches of `run_climate_calibration_cl.jl`: `PIPE2=1` folds the crop pipe into a single circuit; `CO2NIGHT=1` adds dark-hour CO_2 to the likelihood; `SOLARC=1` switches on `solar_cons`, `solar_mass_share`, `par_below`; `CROPTAG` selects the crop input series.

6.4 Equations changed by the flags

Cover node (`use_cover_node`, `solar_split`, `cover_cond`). The cover temperature is the quasi-static solution of a linearised balance,

$$T_c = \frac{k_R\,(T_2 + T_{\mathrm{sky}}) + h_{\mathrm{out}}\,T_{\mathrm{out}} + R_c}{2k_R + h_{\mathrm{out}}}, \qquad k_R = 4\varepsilon_3\sigma\big(\tfrac{1}{2}(T_2 + T_{\mathrm{sky}})\big)^3, \tag{88}$$

with $R_c = a_c I_2$ (`solar_split`) or the cover share of the solar column (`solar_cons`). With `cover_cond` the latent heat of the vapour condensing on the cover is added, $T_c = T_c^{(0)} + \gamma_2\,\mathrm{MV}(T_c)/(2k_R + h_{\mathrm{out}})$, solved by four fixed-point passes, and the vapour balance loses

$$p_{10} = A_c\,[1 - U_1(1 - p_{\mathrm{scr}})]\;6.4\times10^{-9}\;1.7\,|T_2 - T_c|^{0.33}\,\big(V_1 - P_s(T_c)\big)^+. \tag{89}$$

Caveat (audit 21 Oct): the node does not receive the long-wave terms $r_7, r_{11}, r_{12}, r_{13}$ that the inside sends it, so the cover balance does not conserve energy (Section 9).

Heating (`heat_ctrl`).

$$T_{\mathrm{pipe}} = \max\Big(T^{\log}_{\mathrm{pipe}},\; T_2 + \mathrm{clamp}\big(K_p e + K_i\,e^{\mathrm{int}},\,0,\,T_{\max} - T_2\big)\Big), \quad e = U_{11} + \Delta T_h - T_2, \tag{90}$$

with anti-windup on e^{int}. `hc_log_pipe`: where $T^{\log}_{\mathrm{pipe}} > T_2 + 1\,\mathrm{K}$ only the logged pipe heats.

Ventilation (`vent_ctrl`, `vent_split`). The closed-loop opening follows

$$\dot U_8^s = \frac{1}{\tau_v}\Big[\max\big(\mathrm{clamp}(g(T_2 - T^{\mathrm{SP}}_{\mathrm{vent}} - e_0), 0, 1),\; h\,\mathrm{clamp}((\mathrm{HD}^{\mathrm{SP}} - \mathrm{HD})/b, 0, 1)\big) - U_8^s\Big]. \tag{91}$$

In open loop (n58 onward) the logged openings are inputs. With `vent_split`, lee U_l and wind U_w openings give the flow

$$F = \tfrac{1}{2}\Big[U_l^e\sqrt{B + c_l\,n_3 v^2} + g_w U_w^e\sqrt{B + c_w\,n_3 v^2}\Big], \qquad B = \omega_1\nu_6\frac{T_2 - T_{\mathrm{out}}}{T_2 + T_{\mathrm{out}}}, \tag{92}$$

passed to f_7 as the equivalent single opening $U_8 = F/\sqrt{\max(B + n_3 v^2, 0)}$.

Internal mass (`mass_node`, `crop_cap`, `solar_mass_share`).

$$\big(C_m + c_{p,\mathrm{crop}}\mathrm{FW}(t)\big)\dot T_m = h_m(T_2 - T_m) + Q_{\mathrm{sm}}, \qquad Q_{\mathrm{sm}} = \begin{cases} f_m\,I_2 & \text{V2.2 form} \\ f_m\,Q^{\mathrm{sol}}_{\mathrm{floor}} & \texttt{solar_mass_share} \end{cases} \tag{93}$$

and the air loses $h_m(T_2 - T_m)$.

Lamps (`lamp_full`). The full electrical power $Q_l = P_{\mathrm{el}}U_{12}$ is split

$$Q_l^{\mathrm{can}} = f_{\mathrm{can}}\big(1 - e^{-\beta_1 L}\big)Q_l, \qquad Q_l^{\mathrm{air}} = Q_l - Q_l^{\mathrm{can}}, \tag{94}$$

replacing V2.0's $(\eta_{15}+\eta_{16})\alpha_{12}U_{12}$ (air) and $(1-\eta_1)\alpha_2\eta_{14}\alpha_{12}U_{12}$ (canopy). *Caveat:* at $f_{\mathrm{can}} = 0.3$, $L = 2.3$ the canopy heat ($28.1\,\mathrm{W\,m^{-2}}$) is below the absorbed lamp PAR (34.3).

Floor (`floor_full`, `soil_dyn`, `deep_k`).

$$C_{\mathrm{soil}}\dot{T}_5 = k_{\mathrm{ground}}(T_2 - T_5) - k_{fd}(T_5 - T_d) + h_{11} + r_{14}(T_1) - r_{11} + (1-f_m)Q_{\mathrm{floor}}^{\mathrm{sol}}, \qquad \dot{T}_d = (T_5 - T_d)/\tau_d. \tag{95}$$

with $k_{fd} = k_{\mathrm{deep}}$ under `deep_k` (else k_{ground}); the h_{11}, r_{14}, r_{11} terms need `floor_full`, the solar term `solar_split` or `solar_cons`, and the T_d equation `soil_dyn`.

Physiology light (`phys_light`, `photo_lai`, `par_below`). The PAR seen by stomata and photosynthesis is

$$\mathrm{PAR}_{\mathrm{phys}} = \begin{cases} I_9(1 - 0.98U_{\mathrm{black}})(1 - 0.25U_1) + P_{\mathrm{lamp}}U_{12} & \texttt{phys_light} \\ \mathrm{PAR}_{\mathrm{top}} + \mathrm{PAR}_{\mathrm{bot}} + P_{\mathrm{lamp}}U_{12} & \texttt{solar_cons+par_below} \end{cases} \tag{96}$$

and `photo_lai` scales the canopy assimilation by the effective leaf area $(1 - e^{-0.7L})/0.7$.

6.5 The energy-conserving solar column (`solar_cons`)

For each band $b \in \{\mathrm{PAR}, \mathrm{NIR}\}$ with incident $I_b = \eta_b I_2$ the column has four layers and an opaque floor: roof (τ_r, ρ_r), upper screen, lower screen, canopy, floor (reflectance ρ_2 for PAR, $1 - \alpha_7$ for NIR). A screen at closure U has $\tau = 1 - U(1 - \tau_s)$, $\rho = U\rho_s$ (energy screen $\tau_s, \rho_s = 0.75, 0.15$; blackout $0.02, 0.50$; NIR equal to PAR; all *provisional*). The canopy layer is

$$t_c = e^{-kL}, \qquad r_c = \rho_\infty(1 - t_c), \qquad a_c = 1 - t_c - r_c, \tag{97}$$

with $k = \beta_1$, $\rho_\infty = \rho_1$ (PAR) and $k = \beta_3$, $\rho_\infty = 0.35$ (`rho_can_nir`; NIR). The adding method (Vanthoor 2011, eqs. 8.14–8.17) gives the absorbed fractions $A_1, \ldots, A_4$, A_f and the fraction R_{out} returned to the sky, with $\sum A + A_f + R_{\mathrm{out}} = 1$ (checked to 3×10^{-14}). The roof reflectance is derived so that the roof absorbs η_1 (structure, booked to the air) plus a_c (glass, booked to the cover): $\rho_r = 1 - (1 - \eta_1)\tau_1 - \eta_1 - a_c \approx 0.15$. Screen absorption goes a share `scr_abs_air` $= 0.5$ to the air, the rest to the cover. Canopy heat is $r_1 = I_{\mathrm{PAR}}A_4^{\mathrm{PAR}} + I_{\mathrm{NIR}}A_4^{\mathrm{NIR}}$ (replacing $r_2 + r_3$ and the solar part of r_5); $r_8 = 0$; the floor receives $(1 - f_m)Q_{\mathrm{floor}}^{\mathrm{sol}}$ and the mass $f_m Q_{\mathrm{floor}}^{\mathrm{sol}}$. For a season under the n79 inputs the shares are canopy 0.371, floor 0.139, air 0.118, cover 0.118, sky 0.253.

Floor light in the crop path. The crop calibration uses measured canopy-top PAR. With `par_below` it is multiplied by $g(L) = (D_{\mathrm{top}} + \rho_2 D_{\mathrm{floor}})/\tau_r$ from the same column with screens retracted ($g = 1.55$ at $L = 0.5$, 1.15 at 2.3, 1.03 at 5).

6.6 Crop: Aaron's canopy step (`gen_canopy`)

The crop used since Note 94 is the generic Marcelis (1994) cucumber model (Aaron Vélez's port, `CUCUMBER_PARAMS_V2`) with Aaron's canopy extension, run daily per $\mathrm{m^2}$ floor with 3.6 stems $\mathrm{m^{-2}}$. States: leaf DW, stem and root DW, LAI, retained leaves per stem, node progress, fruit cohorts (stage, mass, number), reserve. Daily order:

1. maintenance $R_m = Q_{10}^{(T-25)/10}\sum_o k_{m,o}W_o$; supply $\Delta DM = \max(P_g - R_m, 0)/1.4$;
2. potentials of the start-of-day inventory: vegetative demand $V(T)$·stems and fruit sinks (M94 Richards shape, amplitude set by W_{pot}, clock $1/E_{\mathrm{fruit}}$); allocation proportional to potentials with a carried reserve (M94 eq. 4);

3. vegetative growth: leaf share 0.59; new leaf area = seasonal SLA(day of year) × new leaf DW;
4. development; node appearance $\beta_0 + \beta_1 T$ per stem per day × M94 light factor $(1 - e^{-0.5-0.5\,\mathrm{PAR}})/(1 - e^{-0.5-0.5\cdot 6.8})$; a leaf at every node; flowers from node 5; fruits set = flowers × stems × kept share (grower log) × carbon gate

$$\mathrm{clamp}\Big(\frac{\Delta DM/\mathrm{potential} - r_{\mathrm{low}}}{r_{\mathrm{high}} - r_{\mathrm{low}}}, 0, 1\Big);$$

5. harvest when a fruit reaches 323 g FW (× DMC) or stage 3;
6. pruning to `gen_leaf_target` retained leaves (leaf DW and LAI removed proportionally).

Photosynthesis is the FvCB model of Section 3 on the daily measured T, CO_2 and PAR (with `par_below`, $\times g(L)$). Crop flags in use: `gen_canopy`=1, `gen_parlight`=1, `gen_zclamp`=1 (sink held at stage 1, picked by size), `gen_clampT`=0, multipliers `gen_A`, `gen_V`, `gen_beta`, `gen_kappa` = 1.

Coupling. One-way and offline: the crop is calibrated on measured climate, and `crop_posterior_series.jl` writes the daily LAI, crop respiration at 20 °C and standing fresh mass used as climate inputs (L in r_1, q_1, A; `co2_resp`; `crop_cap`).

7 Best model along the way and recommended configuration

set-up (Notes)	added	result / status
Climate		
n36–n39 (35–40)	closed-loop vents, PI heating, stomata/vapour fixes	limit cycle removed (n37); superseded
n46–n54 (46–54)	closed floor balance, solar split, internal mass, full lamp heat, both pipes	mass on its caps; superseded
n58 (58)	open loop with logged lee/wind vents	bimodal; superseded
n61 (61–62)	air heat-capacity multiplier, sensor term `s_rad`, season-end TRAIN block	**best fit so far**: hourly RMSE 1.30 (all) / 1.20 K (common test) as compared; physical air alone 1.56 / 1.27 K, day −1.24 K. Relies on a measurement term, now excluded
n79 (78–96)	a-priori geometry, cover condensation, crop respiration, dew, shared and stack leakage, R_d Q10, crop heat capacity, dark CO_2 observed; no bias terms	1-h test (outdated config, not converged): TRAIN 1.54, VAL 1.23, TEST 1.50, common 1.66 K
n79 + SOLARC (101–106)	conserving solar column, mass share, floor light	smoke at prior centre: budget closes; −1.2 K (day −1.6, night −1.0), RH ≈ −10 %. **Recommended structure**; no full run yet
Crop		
n94 (94–102)	canopy step, 3.6 stems, fruit numbers observed	converged: yield 34.6/34.0 vs 37.3 kg m^{-2}; fruits 102/101 vs 116
n105 (105–108)	n94 + floor light	yield 35.8/36.2; fruits 107/108; bimodal (small vs large fruit); data prefer the large-fruit mode (seed 2)

Recommended configuration. Climate: `PRIORS=n79 SOLARC=1 PIPE2=1 CO2NIGHT=1`, i.e. the "n79" column of Section 6.3, with the fixed values of Section 8.2. Crop: n105 (large-fruit mode). These are the baseline for the physics work; free parameters stay at their prior centres until a converged n79+SOLARC run exists.

8 Parameter-inference scheme

Both calibrations use the t-walk MCMC sampler with uniform priors [min, max] (a centre is used for the start), a Gaussian likelihood with inferred noise per observable,

$$U(\theta, \sigma) = \sum_k \Big[n_k \log \sigma_k + \frac{\mathrm{SSE}_k(\theta)}{2\sigma_k^2} \Big], \tag{98}$$

two or more seeds, split-$\hat{R}$, explicit burn-in, and disjoint TRAIN/VAL/TEST periods.

8.1 Crop (`calibrate_canopy_growth.jl`)

Driven by measured AiCU air T, CO_2 and canopy PAR ($\mathrm{PAR}_{\mathrm{out}} \times 0.65 \times$ screen factors + 187 μmol lamps). Observations: weekly harvest, weekly harvested fruit number, leaves per stem, fruit development time.

group	parameter	prior [min, centre, max]
M94	W_{pot} (g DM)	20, 60.7, 120
	E_{fruit} (°C d)	150, 275, 450
	β_1 (appearance)	0.05, 0.0882, 0.13
	V_{20} (g stem^{-1} d^{-1})	1, 3.2, 8
maintenance	$k_{m,\mathrm{leaf}}$, $k_{m,\mathrm{stem}}$, $k_{m,\mathrm{fruit}}$	0.01–0.06, 0.005–0.03, 0.003–0.02
canopy	leaf target; N_0	20–30; 2–8
fruit set	r_{low}; r_{high}	0–1; 0.2–2.5
quality	$\mathrm{DMC}_{\mathrm{fruit}}$	0.025–0.045
photosynthesis	J_{max}	$1–3.5 \times 10^{-4}$
noise	$\sigma_y, \sigma_{\mathrm{leaf}}, \sigma_{fd}, \sigma_n$	

Fixed a priori: stem density 3.6, transmission 0.65, harvest size 323 g FW, transplant 14 Aug (LAI 0.25), seasonal SLA, leaf share 0.59, flowers from node 5, M94 sink shape, construction cost 1.4, $Q_{10} = 2$, kept share from the grower log.

8.2 Climate (`run_climate_calibration_cl.jl`, `PRIORS=n79`)

Seven-day blocks with one spin-up day and measured initial state. TRAIN days 7–13, 28–34, 63–69, 91–97, 110–114; VAL 42–48, 77–83; TEST the rest. Observations: hourly means of air T and RH, dark-hour CO_2. Vents are logged inputs (open loop).

sub-model	parameter	prior [min, centre, max]
ventilation	ν_1; ν_2	0.3–0.65–1; 0.005–0.05–0.3
	vs_exp; vs_gwind; vs_cwind	0.5–0.9–1.2; 0.5–1.5–4; 0.5–1–10
	vent_scr_perm	0–0.4–1
leakage	ν_4; leak_cs	10^{-5}–4.5×10^{-5}–10^{-4}; 0–0.3–1
stomata	γ_4 (s m^{-1})	40–82–200
lamps	lamp_fcan	0.05–0.3–0.9
floor	k_{ground}; C_{soil}; k_{deep}	1–8–40; 1.5×10^5–6×10^5–2×10^6; 0.5–1.5–3
mass	C_m; h_m; f_m	2×10^4–5×10^4–1.2×10^5; 1–5–40; 0–0.3–1
cover	h_{out}	10–25–50
fast mass	air_cap_mult	1–3–12
noise	$\sigma_T, \sigma_{RH}, \sigma_C$	0.2–6 K; 0.5–25 %; 10–400 mg m^{-3}

Fixed a priori (compartment documents): air height $\phi_2 = 5.4$ m, $c_p = 1005$, vent area 0.30 of the floor, cover area 1.6, rail pipe 1.0 m m^{-2} of 51 mm (one circuit), PAR transmission 0.65 ($\tau_1 = 0.65/0.9$), CO_2 valve 12.8 g m^{-2} h^{-1}, lamp 117 W m^{-2}, crop specific heat 3800, screen optics, controller gains, $\tau_d = 14$ d. Removed: `s_rad`, `dday`, ΔT_h, lamp screen loss.

8.3 What to infer and what to fix

1. **Fix** what is measurable or documented: geometry, material properties, actuator capacities and spectra, crop counts and dates. Missing values of this kind are questions for the organisers or data sheets.
2. **Infer** only coefficients of approximate sub-models whose form is known: discharge and wind coefficients, leakage, minimum stomatal resistance, convective coefficients, effective storages, crop allocation and development constants, noise levels.
3. **No bias or measurement-offset terms**; a fit that needs one points to missing physics.
4. As the physics becomes descriptive, retire effective parameters: $h_{\text{out}} \to 2.8 + 1.2v$ with a conserving cover; air_cap_mult → explicit fast masses; lamp_fcan → lamp spectrum through the column; vent_scr_perm → screen permeability of a top compartment.
5. Parameters on their bounds (n79 test: γ_4, h_{out} low; ν_4, leak_cs, C_{soil}, k_{deep} high) are structural warnings.

9 Known structural gaps

From the read-only audit of 21 October and the notes:

1. **Cover node not conserving**: it does not receive $r_7, r_{11}, r_{12}, r_{13}$ or pipe radiation ($40\,\mathrm{W\,m^{-2}}$ residual on a test night); r_{11} contains the PAR transmission τ_1.
2. **No top compartment**: screen at air temperature, r_{13} without a screen factor, condensation sharing the crop air. Geometry: screen $\approx 4.5\,\mathrm{m}$, ridge $\approx 5.8\,\mathrm{m}$, $\approx 0.9\,\mathrm{m^3\,m^{-2}}$ above the screen.
3. **Long-wave network**: canopy–floor r_{14} without canopy fraction; NIR extinction (0.27) used for long-wave instead of $K_{\text{FIR}} = 0.94$; pipes radiate only to the canopy (area 0.30 vs geometric 0.16).
4. **Lamps**: canopy heat below absorbed lamp PAR; no lamp light to the floor.
5. **Cold night bias** ($\approx 1.2\,\mathrm{K}$) and dry air (RH $-10\,\%$) at the n79+SOLARC prior centre.
6. Big-leaf photosynthesis; no fruit-abortion model; provisional screen and roof optics; one-way crop coupling.

A Climate-model constants

Physical parameters of the climate subsystem (Section 2). Values as configured in the reference twin; symbols follow the Vanthoor convention ($\alpha_i, \beta_i, \ldots$).

Name	Value	Units	Description
alpha1	1500	J * K^{}-1 * m^{}-2	Heat capacity of one square meter of the canopy
alpha12	74	W * m^{}-2	Total lamp radiation per square meter
alpha2	0.35		Global NIR absorption coefficient of the canopy
alpha3	0.3	m^{}2*m^{}-2	Surface of the heating pipe
alpha4	5	W * m^{}-2 * K^{}-1	Convection heat exchange coefficient of canopy leaf to greenhouse air
alpha5	1500	J * K^{}-1 * kg^{}-1	Specific heat capacity of greenhouse air
alpha6	10000	m^{}2	Greenhouse floor surface area
alpha7	0.5		Global NIR absorption coefficient of the floor
alpha8	1		PAR absorption coefficient of the cover

Name	Value	Units	Description
alpha9	1		NIR absorption coefficient of the cover
beta1	0.7		Canopy extinction coefficient for PAR radiation
beta2	0.7		Extinction coefficient for PAR radiation reflected from the floor to the canopy
beta3	0.27		Canopy extinction coefficient for NIR radiation
C_soil	150000	J m^{}-2 K^{}-1	Heat capacity per floor area of the active soil/floor layer (thermal mass)
delta1	5	W * m^{}-2	Radiation above the canopy that defines sunrise and sunset
delta2	4.3	W * m^{}-2	Empirically determined parameter
delta3	0.54	W * m^{}-2	Empirically determined parameter
delta4	6.1e-07	ppm^{}-2	Coefficient of the CO2 transpiration in the day
delta5	1.1e-11	ppm^{}-2	Coefficient of the CO2 transpiration in the night
delta6	4.3e-06	Pa^{}-2	Coefficient of the vapour pressure in the day
delta7	5.2e-06	Pa^{}-2	Coefficient of the vapour pressure in the night
eps_screen	0.4		Energy-screen FIR emissivity to sky (aluminized/low-e). Separate from epsil5 so it only affects the screen-sky term r12 when the screen is deployed (U1¿0); U1=0 on the venting calibration day, so calibration-safe.
epsil1	0.88		FIR emission coefficient of the heating pipe
epsil2	1		Canopy FIR emission coefficient
epsil3	1		Sky FIR emission coefficient
epsil4	1		Floor FIR emission coefficient
epsil5	1		Thermal screen FIR emission coefficient
epsil6	0.44		External cover FIR emission coefficient
eta1	0.1		Proportion of global radiation that is absorbed by greenhouse building elements
eta10	0		Shadow effect on the discharge coefficient
eta11	0		Effect of shadow on the global wind pressure coefficient
eta12	4.43e-08	kg_vapour * J^{}-1	Amount of vapor that is released when a joule of sensible energy is produced by the direct air heater
eta13	0.057	mg_CO2 * J^{}-1	Amount of CO2 that is released when a joule of sensible energy is produced by the direct air heater
eta14	0.18		Percentage lamps radiation that is NIR
eta15	0.0074		Porcentaje de la radiacion de las lamparas que es calor en el reflector

Name	Value	Units	Description
eta16	0.39		Porcentaje de la radiacion de las lamparas que es calor directo al aire del invernadero
eta17	0.3626		Percentage lamps radiation that is PAR
eta2	0.5		Ratio between PAR radiation and external global radiation
eta3	0.5		Ratio between NIR radiation and global external radiation
eta4	0.554	ppm * mgˆ{}-1 * mˆ{}3	Conversion factor for CO2 of mg*m**-3 to ppm
eta5	0.5		Fan-pad system efficiency
eta6	0.551		Ventilation power reduction factor
eta7	0.8		Ratio between ceiling ventilation area and total ventilation area
eta8	0.9		Ratio between ceiling and total ventilation area, if there is no chimney effect
gamma	65.8	Pa * Kˆ{}-1	Psychometric constan
gamma1	1.25	m * mˆ{}-2	Length of the heating pipe
gamma2	2.45e+06	J * kg_waterˆ{}-1	Latent heat of water evaporation
gamma3	275	s * mˆ{}-1	Strength of boundary layer of canopy for vapor transport
gamma4	82	s * mˆ{}-1	Minimum stomatal resistance of the canopy
gamma5	-1	W * mˆ{}-2	Inverse of slope of the differentiable switch for the stomatal resistance model
h_soil	4	W mˆ{}-2 Kˆ{}-1	Effective air¡-¿floor convective heat-exchange coefficient (thermal-mass buffer)
HEAT_PIPE	363.15		Temperatura del tubo de calentamiento
k_deep	2	W mˆ{}-2 Kˆ{}-1	Conductance from the active floor layer to the fixed deep-soil temperature
k_ground	4.33	W mˆ{}-2 Kˆ{}-1	Effective air¡-¿ground conductance via the floor thermal-mass state (single, identifiable)
lamb1	0		Performance coefficient of the mechanical cooling system
lamb2	0	W	Electrical capacity of the mechanical cooling system
lamb3	1	W * mˆ{}-2 * Kˆ{}-1	Convective heat exchange coefficient between soil and greenhouse air
lamb4	500000	W	Heat capacity of direct air heater
lamb5	18000	mˆ{}2	Cover surface
lamb6	2.8	W * m_coverˆ{}-2 * Kˆ{}-1	Variable of heat exchange by convection between the roof and the outside air
lamb7	1	J * mˆ{}-3 * Kˆ{}-1	Variable of heat exchange by convection between the roof and the outside air
lamb8	2		Variable of heat exchange by convection between the roof and the outside air
lamb9	0.05		eficiencia del intercambio de calor entre las capas de aire

Name	Value	Units	Description
n_pipes	2		Numero de tuberias de calentamiento
nu1	1		Shadowless discharge coefficient
nu2	0.1		Global wind pressure coefficient without shadow
nu3	2352.9	m^{}2	Side surface of the greenhouse
nu4	0.0001		Heat leakage coefficient
nu4CO2	0.0001		CO2 leakage coefficient
nu5	2000	m^{}2	Maximum ceiling ventilation area
nu6	1	m	Vertical dimension of a single open respirator
nu7	0.85	W * m^{}-1 * K^{}-1	Soil thermal conductivity
nu8	0.64	m	Floor to ground distance
omega1	9.81	m * s^{}-2	Gravity acceleration constant
omega2	8314	J * kmol^{}-1 * K^{}-1	Molar gas constant
omega3	0.0075		Percentage of CO2 absorbed by the canopy
phi1	0.051	m	External diameter of the heating pipe
phi2	4	m	Average height of greenhouse air
phi3	28.9647	kg * kmol^{}-1	Masa molar del aire
phi4	0	metros	Altitud del invernadero
phi5	0.014	kg_water * kg_air^{}-1	Water vapor contained in the fan-pad system
phi6	0.0079	kg_water * kg_air^{}-1	Water vapor contained in the outside air
phi7	16.7	m^{}3 * s^{}-1	Capacity of air flow through the pad
phi8	666.6	m^{}3 * s^{}-1	Air flow capacity of forced ventilation system
phi9	0.916	kg * s^{}-1	Fog system capacity
psi1	18	kg * kmol^{}-1	Molar mass of water
psi2	13300	mg * s^{}-1	Capacity of the external CO2 source
psi3	30.031	g * mol_CH2O^{}-1	Molar mass of the CH2O
rho1	0.07		PAR reflection coefficient of the cannopy
rho2	0.65		Floor reflection coefficient PAR
rho3	1.2	kg * m^{}-3	Air density
rho4	1.2	kg * m^{}-3	Air density see level
sigma	5.67e-08	W * m^{}-2 * K^{}-4	Stefan-Boltzmann constant
tau1	0.77		PAR transmission coefficient of the Cover
tau2	1		FIR transmission coefficient of the Cover
tau3	0.11		FIR transmission coefficient of the thermal screen

B Photosynthesis constants

Parameters of the FvCB leaf model and its temperature responses (Section 3).

Name	Value	Units	Description
ab	0.85		Leaf absorbance
alpha	0.35		Product of the leaf absrovance, the fraction of light directed to PSII and the maximum quantum effiuciency of PSII; initial slope of an J/Iin curve
C_ev1	4.97685e-11	mol_phot * mˆ{}-2 * sˆ{}-1	Constant in the formula of f_R
C_ev2	6.25e-12	mol_phot * mˆ{}-2 * sˆ{}-1	Constant in the formula of f_R
C_ev3d	0.61	mol_air * mol_CO2ˆ{}-1	Constant in the formula of f_C
C_ev3n	1.1e-05	mol_air * mol_CO2ˆ{}-1	Constant in the formula of f_C
C_ev4d	4.3e-06	Paˆ{}-2	Constant in the formula of f_C
C_ev4n	5.2e-06	Paˆ{}-2	Constant in the formula of f_C
E_Soc	-24460	kg * mˆ{}2 * sˆ{}-2 * molˆ{}-1	Sco activation energy
f	0.15		Correction factor for the spectral quality of the light
fc	2.95331e+10	g * s * mol_CO2ˆ{}-1	Factor to transform μmol_CO2/sec to grams_CH20/day
J_max	0.0002	mol_phot * mˆ{}-2 * sˆ{}-1	Maximum electron transport rate
k_Ag	1e-06	mˆ{}3 * gˆ{}-1 * sˆ{}-1 * mol_CO2 * mol_airˆ{}-1	Constant for units transformation
K_C25	0.0003	mol_CO2 * mol_airˆ{}-1	Michaelis-Menten constant for CO2
k_d	86400		Factor to transform sˆ{}-1 into dˆ{}-1
k_fc	1e-06	mol_CO2 * mol_airˆ{}-1	Constant for units completion
k_JV	1		Auxiliary constant which transforms the units of the electron transport rate J to those of the maximum Rubisco rate V_cmax
K_O25	0.3	mol_O2 * mol_airˆ{}-1	Michaelis-Menten constant for O2
k_T	1	C	Auxiliary constant to add temperature units
ks	0.5		Stomatal ratio
O_a	0.21	mol_O2 * mol_airˆ{}-1	O2 concentration in the environment
phi	2	mol_O2 * mol_CO2ˆ{}-1	Ratio of oxygenation to carboxylation rates
Q10_KC	2.1		Temperature response of Michaelis-Menten constant for CO2
Q10_KO	1.2		Temperature response of Michaelis-Menten constant for O2
Q10_tau	2.1		Temperature response of specificity factor

Name	Value	Units	Description
Q10_Vcmax	2.4		V_cmax value at 25C
r_m	100	s * mˆ{}-1	Minimal stomatal resistance
Rb	711	s * mˆ{}-1	Stomatal resistance of the canopy boundary layer
Rd_day	1e-06	mol * mˆ{}-2 * sˆ{}-1	Rate of respiratory CO2 released from the mitocondria
Rgas	8.31447	mˆ{}2 * kg * sˆ{}-2 * Kˆ{}-1 * molˆ{}-1	Universal Gas constant
Rs	5e-06	mol_phot * mˆ{}-2 * sˆ{}-1	Radiation setpoint to switch day and night
S	-86400	mˆ{}2 * s * mol_photˆ{}-1	Constant in the formula of Sr
Sco25	2800	Pa * Paˆ{}-1	Rubisco CO2/O2 specificity factor (Sc/o) at 25C
tau_25	2600	mol * molˆ{}-1	Specificity factor
theta	0.7		Empirical curvature factor
V_cmax25	0.00015	mol_CO2 * mˆ{}-2 * sˆ{}-1	Maximum Rubisco Rate, per unit area

C Crop-growth constants

Parameters of the cucumber source–sink growth model, including flowering, fruit and the leaf-cohort canopy (Sections 4).

Name	Value	Units	Description

D State variables, inputs and controls

Table 6: Dynamic state variables (climate ODE).

Symbol	Units	Meaning
T_1	K	Canopy (leaf) temperature
T_2	K	Greenhouse-air temperature (the controlled variable)
C_1	mg m^{-3}	Greenhouse-air CO_2 concentration
V_1	Pa	Greenhouse-air vapour pressure
T_5	K	Floor / soil thermal-mass temperature

Table 7: Inputs to the climate right-hand side (weather drivers and controller-supplied quantities).

Symbol	Units	Meaning
I_1	m^2 m^{-2}	Leaf area index (LAI), from the growth model
I_2	W m^{-2}	Outside global (short-wave) radiation
I_3	K	Heating-pipe temperature (from the heating controller)

Symbol	Units	Meaning
I_4	K	Effective sky temperature (Berdahl–Martin)
I_5	K	Outside air temperature
I_6	K	Mechanical-cooling temperature
I_7	K	Floor temperature seen by the FIR/conduction terms ($= T_5$)
I_8	m s^{-1}	Outside wind speed
I_9	W m^{-2}	PAR reaching the canopy $= (1-\eta_1)\tau_1\eta_2\, I_{docel}$
I_{10}	mg m^{-3}	Outside CO_2 concentration
I_{11}	Pa	Outside vapour pressure
T_{deep}	K	Deep-soil (anchor) temperature, from weather

Table 8: Control actuators U_1–U_{12} (each in $[0,1]$ unless noted).

Symbol	Actuator	Role / entry point
U_1	Thermal / energy screen	FIR terms r_{12}, screen transmission g_2, g_3; deploy/crack
U_2	Fan–pad (evaporative) inlet	h_2, p_2, p_6, o_3 (0 in reference config)
U_3	Mechanical cooling	(0 in reference config)
U_4	Direct air heater	o_1, p_4 (0 in reference config)
U_5	Shade/black-out screen	modifies vent discharge n_1, n_3 (0 in ref.)
U_6	Side (roll-up) ventilation	wind-driven exchange f_5
U_7	Forced ventilation (fans)	f_3 (0 in reference config)
U_8	Roof (ridge) ventilation	buoyancy+wind exchange f_7
U_9	Fogging / misting	p_3 (latent) and l_2
U_{10}	External CO_2 dosing valve	source o_2
U_{11}	(reserved / measurement)	–
U_{12}	Supplemental lamps (on/off)	PAR to canopy; r_5 (NIR), h_{12} (heat)